THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

December 12, 2025 2:30 pm - 5:30 pm

Answer ALL **5 (FIVE)** questions from Section A and B.
Marks for sub-questions (if any) are given.

Use a separate answer book for each section.

Candidates are permitted to bring to the examination ONE sheet(s) of A4-sized paper with printed/written notes on both sides.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

SECTION A

1. (a) As shown in Figure 1(a), the beam ABCDEF is subject to external concentrated force at F equal to 10kN. D is a hinge connection. There are supports at E and C points. Neglect self-weight of the beam. The steel material with square solid section is used for the beam with uniform section size. The maximum flexural stress is limited to no more than 100MPa. Safety factor is 1.0. Neglect shear stress in this problem.

   (i) Draw the bending moment diagram (BDM) with pertinent values. Noe that bending moment should be drawn on the **tensile** side. **(5 marks)**

   (ii) Find the cheapest cross section size to meet the strength requirement. **(5 marks)**

   (b) A plane frame is constructed as shown in Figure 1(b). The beam is fixed to the column at point B. The external loads include 50kN horizontal load at B and Uniformly Distributed Load (UDL) of 5kN/m along BCD beam. A thin-walled box section with size of 500mmx500mm and wall thickness of 5 mm is used for all beams and columns. Neglect self-weight of the structure. Neglect buckling or shear stress for this problem.

   (i) Draw the bending moment diagram (BDM) of the frame with pertinent values. Noe that bending moment should be drawn on the **tensile** side. Note that BMD of all components need to be plotted. **(5 marks)**

   (ii) Find the maximum compressive normal stress of the structure. **(5 marks)**

2. (a) Consider load P acting on the simple beam AB as shown in Figure 2(a). The von Mises criterion is used for the material. The yield strength is obtained from tensile test as 40 MPa. The safety factor is 1.0. Data are as follows: P=10kN, L=40m, d=10m, cross section is shown in Figure 2(a).

   (i) Check the maximum normal stress and maximum shear stress of the beam. **(5 marks)**

   (ii) Check the safety of the beam using von Mises criterion. **(5 marks)**

   (b) A point is subject to plane stress state as shown in Figure 2(b). x axis is horizontal and pointing to the right. y axis is vertical. The magnitude of shear stress is 50 MPa. The direction of shear stress is as shown in Figure 2(b). The von Mises criterion is used for the material. The yield strength is obtained from tensile test as 150 MPa. Safety factor is 1.0.

   (i) Determine the stress acting on an element oriented at an angle θ=60 degrees from the x axis. Rotating counterclockwise from x axis is positive. **(5 marks)**

   (ii) Check the safety of the point using von Mises criterion. **(5 marks)**

3. The horizontal beam ABC shown in the figure is supported by columns BD and CE. The beam is prevented from moving horizontally by the pin support at end A. Each column is pinned at its upper end to the beam, but at the lower ends, support D is a guided support and support E is pinned. Both columns are solid steel bars of square cross section with width equal to 20 mm. Elastic modulus is 200 GPa, Poisson ratio is 0.3. A load Q acts at distance $a=0.5m$ from column BD. What is critical load Q_{cr}. **(10 marks)**

FIGURES

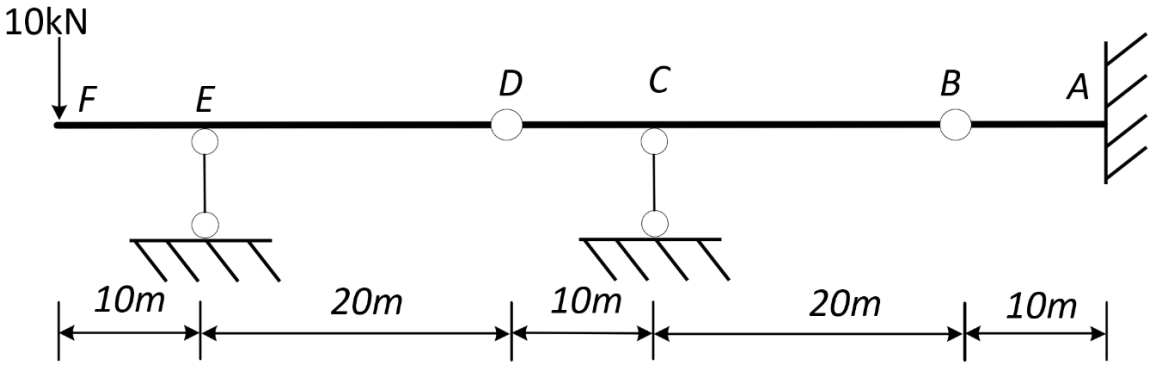


Figure 1(a). Beam

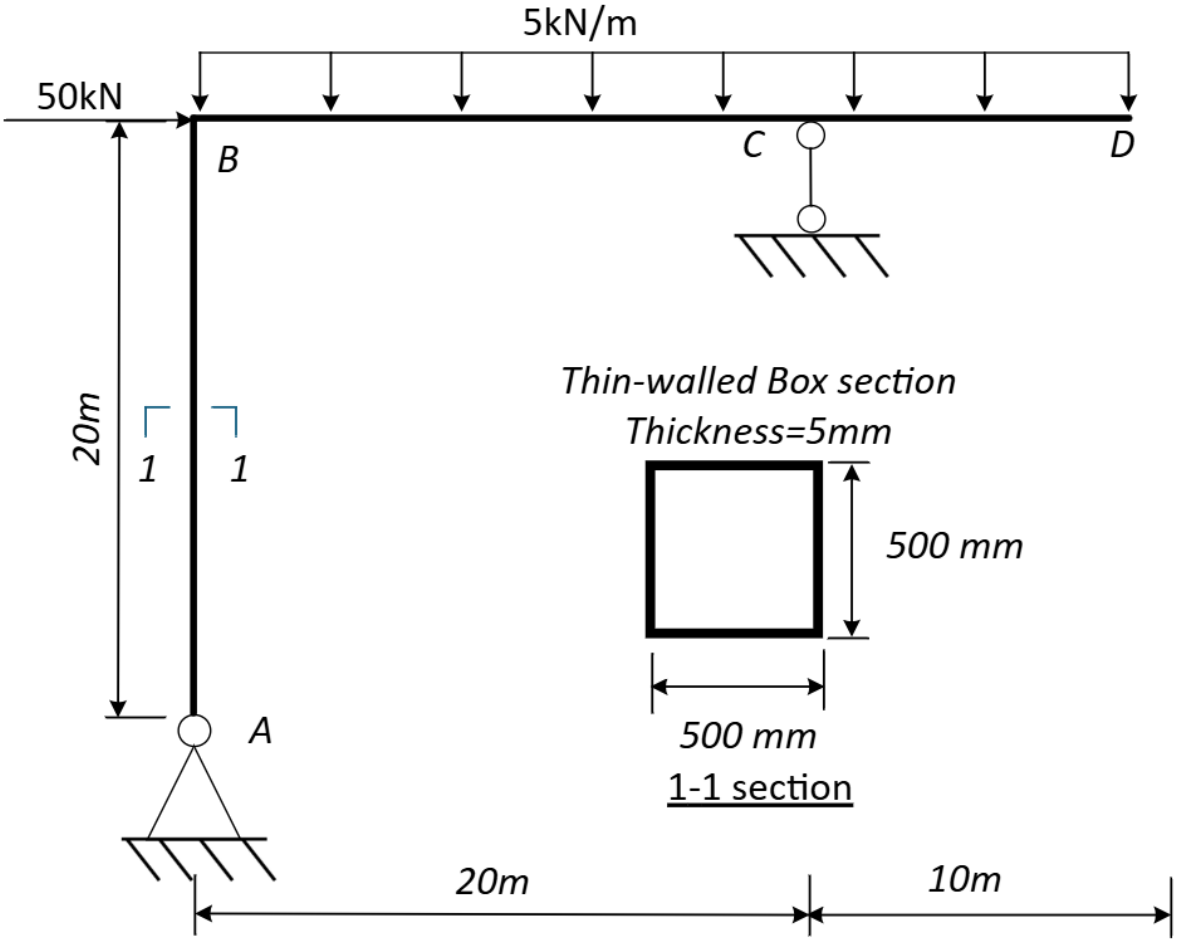
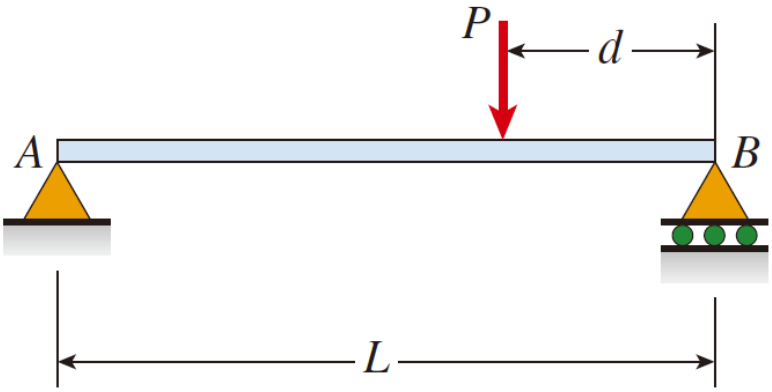
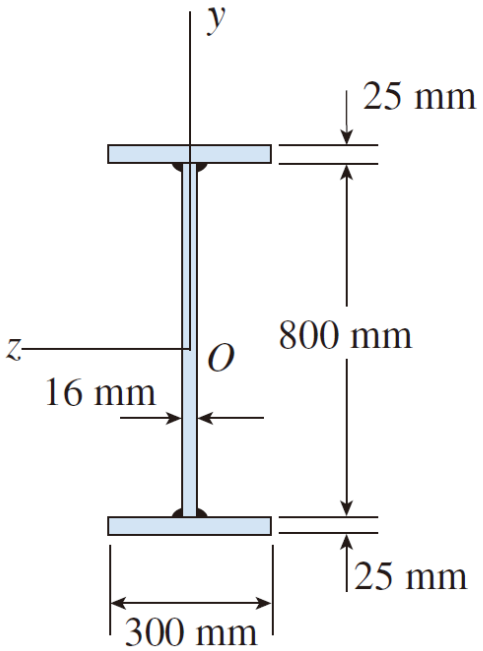


Figure 1(b). Plane frame



Beam loading



Cross-section of beam

Figure 2(a). Simple beam section

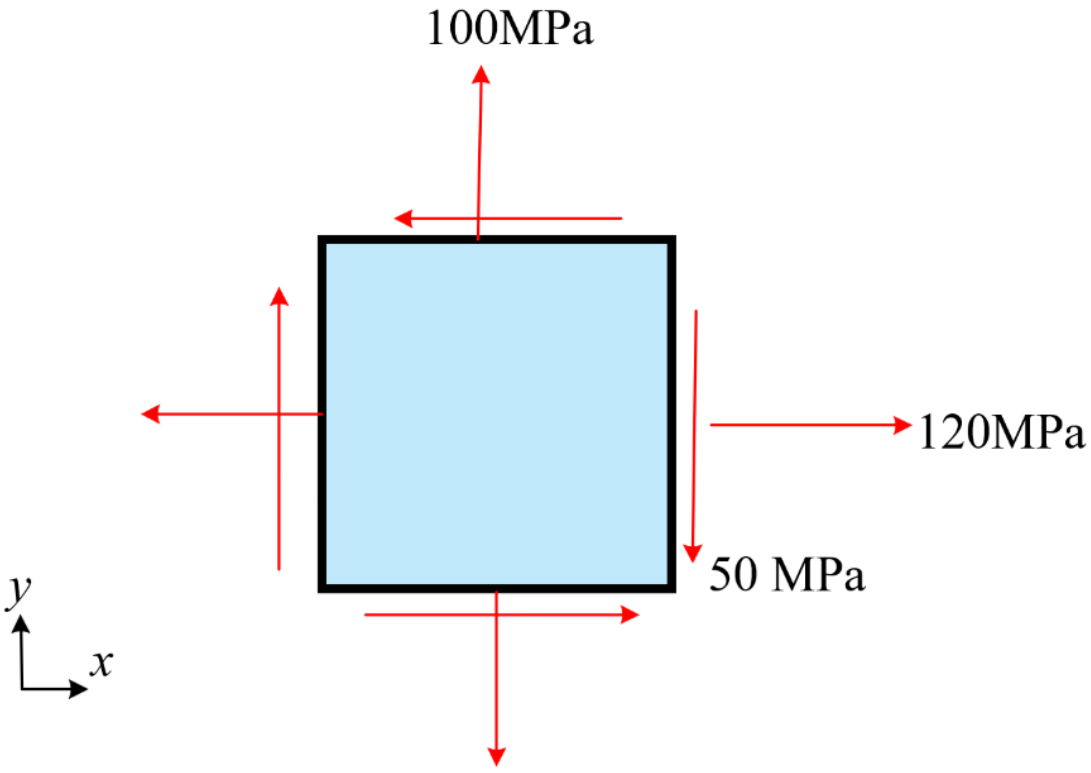


Figure 2(b). Plane stress at a point

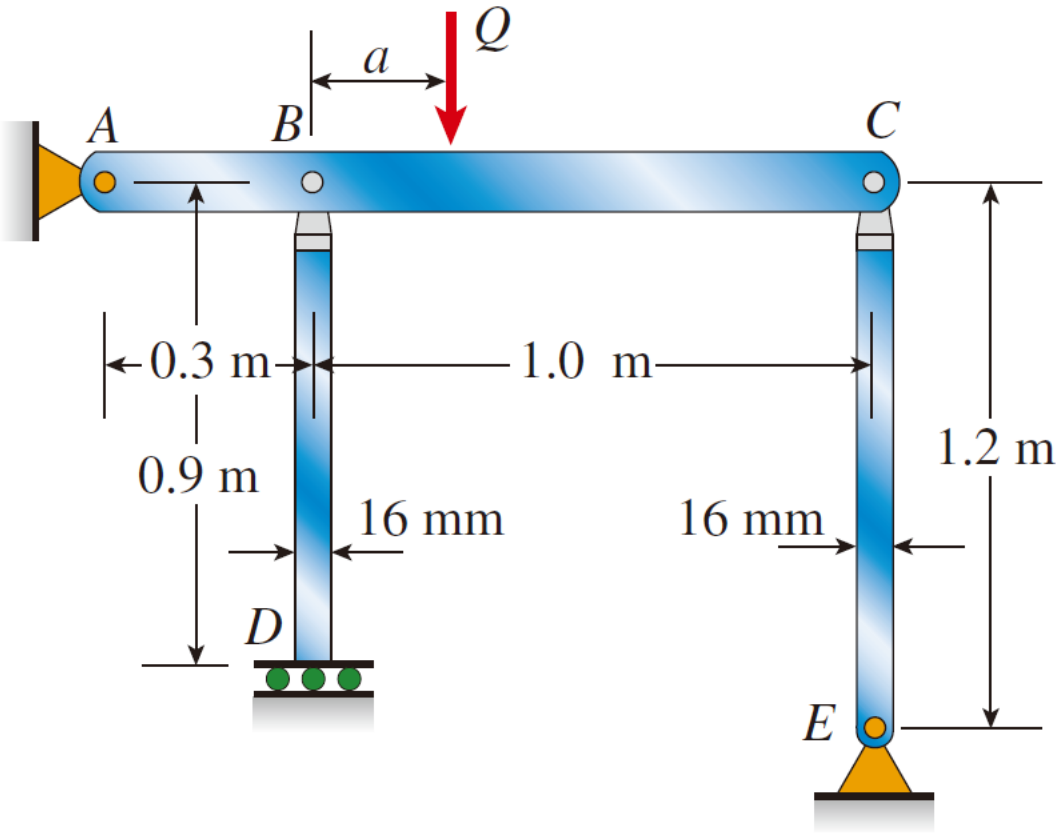


Figure 3. Column

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